

Enhancing Decision-Making of Network Intrusion Analysis assisted by explainable AI for Real-time Security Monitoring

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Abstract—Nowadays, Artificial Intelligence (AI) and Machine Learning (ML) are being used as critical technologies regardless of fields. For network security monitoring, AI/ML techniques also broadly adopted to detect various threats from enormous network traffic. It definitely can enable us to make quicker and more accurate threats detection; however, since AI systems are often operated as black boxes, it is hard to understand the cause of detection without additional analysis. To solve these issues, there have been several attempts using explainable-AI (XAI) techniques to interpret decision processes of AI. Nevertheless, it still has limitations especially in terms of response at real-world Security Operations Centers (SOCs), because it only focuses on revealing an importance of pre-defined features. This paper proposes a practical and security monitoring-response friendly XAI framework, which improves rapid and accurate decision-making by human agents of SOCs. Particularly, the framework provides an intuitive evidence of cyber attacks or anomalies through four steps as follows: 1)extracting semantic features, 2)scoring threats importance, 3)ranking risk influences and 4)visualizing decision evidences. For evaluation purposes, we work with a dataset consisting of real-world network traffic and annotated threats factors by security monitoring experts. Furthermore, experimental results demonstrate the effectiveness of our proposed framework in terms of a decision-making support for real-time security monitoring.

Index Terms—Explainable Artificial Intelligence, Intrusion Detection, Decision-making Support, Security Monitoring, Knowledge Extraction

I. INTRODUCTION

The advancement of network communication has led to the integration of all devices through the networks, resulting in enhanced efficiency and convenience across society. However, this increase in connectivity has also expanded the potential targets for cyber attacks; thus the damage caused by cyber attacks is also increasing steadily and rapidly. To protect critical and essential digital assets against cyber attacks, most institutions and organizations are trying to implement various security devices, including Firewall (FW), Intrusion Detection or Prevention System (IDS/IPS), and other similar equipment. In terms of network security monitoring, however, most of these security devices perform signature-based detection,

which results in a high number of false positive alerts. Therefore, further analysis tasks of human agents are accompanied inevitably to find real threats from amongst huge amount of alerts, even we equipped various security devices [1–3].

In response to the increasing complexity and quantity of cyber threats with regard to the network security, there is a growing interest in utilizing AI to bolster cyber security. The advancement of AI/ML techniques have being applied for detecting various types of threats [4, 5], as well as to automate manual tasks inside processes of real-world SOCs [6–8]. However, while AI has demonstrated substantial benefits across various security domains, it is hard to fully contribute the power of AI/ML techniques. Because even minor errors or inaccurate classifications in SOCs can lead to irretrievable damages. Considering the task of SOCs especially, the AI/ML algorithm only provides the result of threats detection, thus human agents difficult to understand a reason of detection and handle the treats with proper strategies. For this reason, it is really hard to prove AI/ML's real worth and only used restrictively as supplementary tools in real-world SOCs.

Under these circumstances, XAI is rapidly emerging as a key technique to accelerate a usage of AI/ML into the SOC environment. Since XAI has ability to provide evidences and reasons of AI/ML's decision [9–11], it can make human agents easy to identify a critical causes of threat. Particularly, significant factors in a network packet can be explained as a feature [12], which involves malicious/suspicious levels from the result of threats detection or classification [13, 14].

However, there are significant limitations to utilize these works in the real-world security monitoring. The existing XAI approaches in network security [12, 13, 15] extract features from packet header information, such as five-tuple or statistics. The explanations as the feature importance can provide an outline of threats, but it is impossible to figure out a threats behavior; because it is usually involved in a content from network payloads. On the other hand, there are attempt to utilize payload contents as a target of explanations [14, 16]. Although these approaches are friendly to human agents in terms of identifying a causes of threats, it still requires additional analysis to understand an implicit causality;

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because the XAI result do not consider semantics in respect of human understandable languages (e.g., incomplete word, partial machine code and so on). Consequently, this gives rise to the further cost of time and labor in the real-world SOC, where demands rapid decision making and response against threats.

In order to combat the problem as practical use of XAI, this paper proposes a framework that provides intuitive and human-friendly XAI interpretations to support decision making from cyber threats. For this purpose, we firstly define a tokenizing method that split contents of payload as a feature considering semantic characteristics (**Extracting**). Next, the feature is calculated by Shapley value [17] (**Scoring**) and ranked with account of a consistency and practical influence of each feature value (**Ranking**). The ranked features finally are visually indicated by coloured highlighting according to the rank, in order to support intuitive decision making from human agents (**Visualizing**).

For evaluation purposes, especially, we construct a dataset consisting of real-world network traffic (i.e., threats alert from IDS/IPS) with threats factors (i.e., tagged text in payload) annotated by security monitoring experts. By using this dataset, we try to validate the effectiveness of our proposed framework in terms of supporting decision making in real-world SOC. Our main contributions are summarized as follows:

- We propose an explanation framework with novel approaches, i.e., Extracting, Scoring, Ranking and Visualizing, to explain what factors are an evidence of threats so as to assist decision making of human agents.
- We conduct experiments and evaluations on real-world datasets to demonstrate the effectiveness of proposed framework for not only identifying evidence of threats, but also supporting intuitive decision making.

This paper is structured as follows: Section 2 covers brief introduction of XAI technology and related XAI researches on in security field. Section 3 provides a detailed explanation of the proposed framework with four novel approaches. Section 4 experiments and evaluates the proposed framework on real-world datasets. Finally, Section 5 discusses the significance and limitations of this study, and concludes this paper with future works.

II. RELATED WORK

To mitigate the black-box nature of AI by providing the rationale behind AI predictions, various Explainable AI (XAI) techniques have been extensively researched. This chapter will discuss well-known XAI approaches and the interpretability they propose. Furthermore, it will introduce related studies that have attempted to integrate XAI into the field of cybersecurity.

A. Explainable Artificial Intelligence(XAI)

Representative XAI techniques that have been studied include SHAP[17] and LIME[18], whose approaches involve calculating and providing the impact of individual features. These studies offer interpretations based on embedded features and provide the influence of each feature on the AI model's

decisions, thereby allowing for the examination of the causes and reasons behind AI's decisions.

In other words, analysts can understand the learning trajectory of the AI model by using the global interpretation results that provide the overall feature impact generated by XAI. Additionally, local interpretation results that provide the feature impact on individual data points allow analysts to understand the reasoning behind a single AI decision.

Thus, using XAI, it is possible to review the justification of the current AI's learning and predictions, thereby mitigating the black-box characteristics of AI. Through such effects, current XAI techniques are being used as key methods for decision support by analysts and as foundational research for continuous model improvement by understanding the reasoning behind the decisions of operational AI models.

B. XAI in Cyber Security

The adoption of XAI is essential in the field of security. The volume of daily traffic in cybersecurity has reached a point where it is challenging for analysts to handle it alone, making the introduction of AI crucial. However, in the security field, where even minor mistakes can escalate into major incidents and the risk of false positives is high, it is necessary to provide a valid rationale for AI's classification and prediction results.

If is used only AI to present simple classification and prediction results, security experts will find it difficult to trust AI's decisions due to its black-box nature and formulate response strategies based on them. This situation leads to a decrease in the usability of AI models. Moreover, in the ever-changing security environment where attack techniques frequently evolve, if the reasons behind true positive and false positive cases, which inevitably occur when using AI, are not understood, continuous model improvement becomes impossible. Therefore, to effectively utilize AI in the security field and address these issues, the introduction of XAI is essential.

Through such studies [12, 13], by providing the basis for AI's decisions and further offering the trend of learning, analysts can understand the reasons behind AI's judgments and anticipate its effective utilization. Additionally, by integrating these XAI techniques into security, security analysts can improve the reliability of AI models by receiving plausible reasons for attack classifications based on key features.

However, numerous challenges exist in introducing this into real-world SOC environments. Previous studies [14–16] primarily focus on providing appropriate post-hoc explanations for AI decisions to verify their validity. In the security domain, where the risk of false positives is high, fully delegating attack judgments to AI involves significant risks. Consequently, decision-making still relies on security experts, and their review activities remain essential. Therefore, decision support technologies that enhance the accuracy and speed of security analysts' evaluations are crucial in network security monitoring.

In this paper, we initially extract intuitive evidence of attacks or anomalies within network packets based on real data

collected from SOC environments. Subsequently, we calculate the influence of the extracted evidence through XAI and identify the key contributors to suspicion judgments using the proposed scoring method. Finally, we propose a practical XAI framework that clearly presents this information to security experts, thereby supporting decision-making.

III. PROPOSED FRAMEWORK

This chapter introduces a framework for providing effective decision support to security analysts in real-world environments. As shown in Figure 1, The proposed framework has four main phases. Through it, analysts identify critical threat factors present in anomalous payloads and provide intuitive interpretability and decision support through a ranking-based visualization.

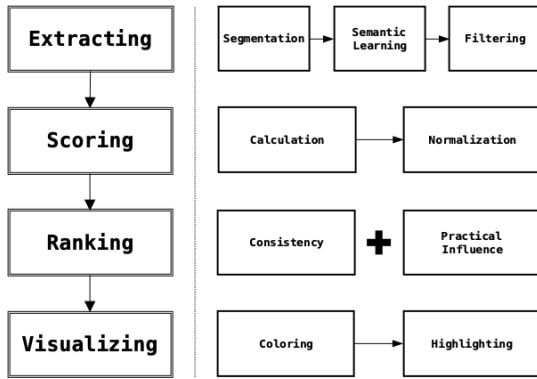


Fig. 1: The Architecture of Proposed Framework

A. Extraction Semantic Features

In the first phase of proposed framework is **Extracting**. In this phase, we propose a method for extracting semantic features by focusing on raw web data. The payload of real-world packets is composed of Natural Language (NL) and Natural Language Processing (NLP) is required to convert NL into vectors to learn these. We used the SentencePiece tokenizer [19] as a natural language processing technology. This technology can automatically tokenize meaningful words by giving high scores to words that frequently appear in the vocabulary of the collected original data.

In particular, network behavior information is contained in the payload, and an example of the HTTP payload is shown in Figure 2. As you can see in the example, the payload consists

```

Method Request_URL Request_Query Version
Start-Line GET /search.naver?query=ChatGPT&where=nexsearch&sm=top_hity&fbm=1 HTTP/1.1
Host: search.naver.com
Referer: https://www.naver.com/
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64 (KHTML, like Gecko))
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9,ko;q=0.8
Empty-Line

```

Fig. 2: Example of Payload from Real-world Traffic

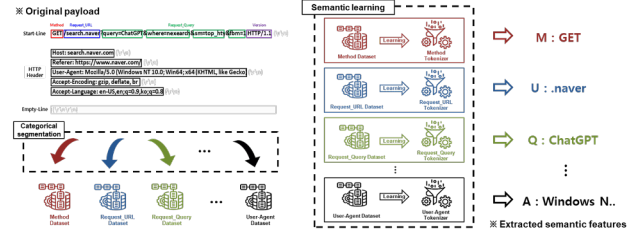


Fig. 3: Concept of proposed Semantic Learning

of several categories, such as Request_Query, Request_URL, User-Agent, etc. These categories have a structure that is segmented based on specific delimiters such as '/', '?', and '&'. In particular, the network behavioral values that appear between the same categories are similar, but the values that appear between different categories are very different.

Therefore, as shown in Figure 3, we propose **Semantic Learning**, which splits the original payload into categories and then trains a category-specific tokenizer, instead of pre-processing the network payload with a single tokenizer.

$$Original_Raw = Category_1, Category_2, \dots, Category_k$$

$$Result_Token_i = Tokenizer_i(Category_i) \quad (1 \leq i \leq k)$$

$$Feature\ F_c = \sum_{i=1}^k Result_Token_i \quad (c = category) \quad (1)$$

Thus, by applying Equation (1), where k represents the number of categories, embedding information for each of the k categories is aggregated into the feature, F_c . The above allows each tokenizer to focus on network behaviors that appear in a specific category, thereby extracting more understandable features.

B. Scoring Threats Importance

The second phase of the proposed framework is **Scoring**. In this phase, the influence for the extracted features F_c is calculated using SHAP[17]. Applying this technique as shown in Equation (2), assigns a Shapley value to each data, which allows us to see how much influence each data had on the AI's prediction. By aggregating all the calculated Shapley values and presenting them, we can obtain a global explanation result that illustrates the general influence trend of each feature.

$$Shapley_values = \{\phi_1, \phi_2, \dots, \phi_m\}$$

$$\phi_i = \sum_{S \subseteq F - \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} (f(S \cup \{i\}) - f(S)) \quad (2)$$

Similar to approaches in previous studies[14], identifying significant features through global interpretation, provides the possibility of a clear interpretation using only the top features, without the need to compare all features.

C. Ranking Risk Influences

The third phase of the proposed framework is **Ranking**. In the traditional ranking approach calculates the influence

Algorithm 1 Scoring and Ranking of Threats Feature

Input: Dataset $S = \{D_1, D_2, \dots, D_n\}$,

Data $D_k = \{t_k^1, t_k^2, \dots, t_k^m\} (1 \leq k \leq n)$,

Trained model M

Output: Feature Scores List $SL = \{s_1, s_2, \dots, s_m\}$

```

1:  $e = \text{Explainer}(M)$ 
2: for  $i \leftarrow 0$  to  $m$  do
3:    $POS_i = [], NEG_i = []$ 

4:   for  $j \leftarrow 0$  to  $n$  do
5:      $sv_j^i \leftarrow e(cv_j^i), cv_j^i \leftarrow \text{freq}(t_j^i) \text{ in } D_j$ 
6:     if  $sv_j^i \geq 0$  then
7:        $POS_i = POS_i \cup \text{MAP}(cv_j^i)$ 
8:     else
9:        $NEG_i = NEG_i \cup \text{MAP}(cv_j^i)$ 

10:    if  $cv_j^i > 0$  then
11:       $Value\_List_i = Value\_List_i \cup sv_j^i$ 

12:     $cons_i = (1 - \text{Entropy}(POS_i)) + (1 - \text{Entropy}(NEG_i))$ 
13:     $prac_i = \text{Avg}(Value\_List_i)$ 

14:     $s_i = cons_i * prac_i$ 
15:     $SL = SL \cup s_i$ 
16: return  $SL$ 
  
```

for each feature by taking the absolute values and averaging them. Therefore, this approach ranks features with the largest overall influence at the top, regardless of whether their influence is positive or negative. However, in cybersecurity, it is essential to detect attacks by identifying the differences between normal and malicious activities. Therefore, in this phase, we proposed new equation based on **Consistency** and **Practical influence** to extract critical threat factors. Through the proposed approach, analysts can more clearly identify the features that have a significant impact on attack detection. The novel ranking method we propose is detailed in Algorithm 1, and the two main concepts of measuring the rank are discussed in more detail in the subsections.

1) *Consistency measurement*: The first concept for the proposed novel ranking is the measurement of consistency. If a feature is important for attack judgment, it should appear only in the attack payload, and when its impact is measured, it should consistently affect only the attack. In other words, to extract critical threat factors, we must also consider the consistency of features.

We can identify whether a feature occurred through its feature value, and we can determine whether the feature contributed to normal prediction or attack prediction through the Shapley value. Based on this assumption, we propose a method for measuring the consistency scores of features as outlined in Algorithm 1. The overall concept of consistency measurement is as shown in Figure 4, and the detailed process will be covered below.

$$\text{MAP}(cv_k) = \begin{cases} 1, & cv_k \geq 1 \\ 0, & cv_k = 0 \end{cases} \quad (3)$$

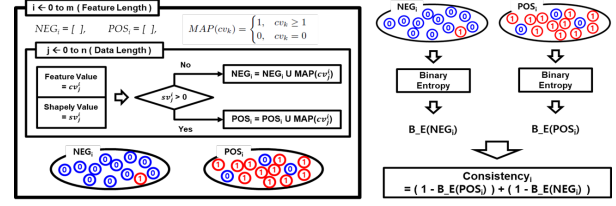


Fig. 4: Concept of consistency measurement

First, for each feature, the data are divided based on the Shapley value of 0 into data contributing to normal, Negative_Data, and data contributing to attacks, Positive_Data. We apply the equation (3) to map whether the current feature has occurred or not to 0 or 1. Through this, as shown in Figure 4, the POS/NEG group of each feature stores whether the feature occurs or not as (0, 1) when the feature affects the attack or normal.

$$\text{Entropy}(x) = -1 * \sum_{i=1}^n p(x_i) * \log(p(x_i)) \quad (4)$$

$$B_E(X) = -1 * [p * \log(p) + (1 - p) * \log(1 - p)]$$

To measure consistency within each group, we measure each feature's consistency based on entropy. Specifically, in accordance with the current SOC scenarios for binary classification of attacks and normal activities, we use binary entropy as described in Equation (4).

$$\text{Consistency}_i = (1 - B_E(pos_i)) + (1 - B_E(neg_i)) \quad (5)$$

However, Binary_Entropy is calculated lower values when attacks and normal are well separated. Therefore, by using '1 - Binary_Entropy(Group)', we ensure that higher consistency results in higher values. To more effectively reflect the overall consistency of normal and attack judgments, we propose a consistency measurement formula that incorporates the entropy of both groups, as shown in Equation (5).

2) *Practical influence measurement*: The second concept for the proposed scoring is measuring the practical influence. The traditional ranking methods use all available data to calculate the average without conditions, which may incorporate the influence of unnecessary data values. Therefore, instead of applying the average to all data, we propose a method to measure actual influence by calculating the influence only for significant data where mapping has occurred. The overall concept of practical influence measurement is as shown in Figure 5, and the detailed process will be covered below. As shown in Figure 5 and Equation (6), where i represents index of feature and j means index of data, in this paper, we apply the average only to Shapley values of data where the actual feature value is 1 or higher, thereby measuring the practical influence of each feature more accurately.

$$Value_List_i = \{cv_j^i \in D_i \mid cv_j^i > 0\}$$

$$\text{Practical_Influence}_i = \text{Avg}(Value_List_i) \quad (6)$$

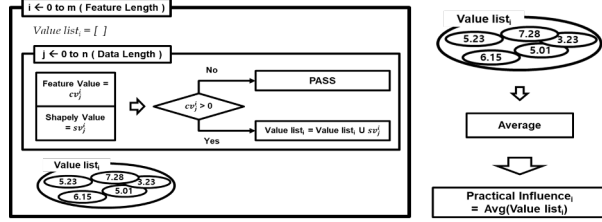


Fig. 5: Concept of practical influence measurement

By performing measurements, such as *Consistency measurement* and *Practical influence measurement*, we derive the consistency score, $Consistency_i$, and the practical influence measure, $Practical_influence_i$, for the i_{th} feature in the dataset. To combine these metrics into a single indicator for the i_{th} feature, we multiply these metrics as shown in Equation (7), resulting in a composite score termed $Score_i$.

$$Score_i = Consistency_i * Practical_Influence_i \quad (7)$$

Calculating the Score for all features in the dataset, as proposed in Algorithm 1, allows features with high consistency and practical influence to yield higher Scores. Subsequently, sorting these features based on their Scores for global interpretation enables the identification of key features that consistently appear in suspected attack data and have a significant influence.

D. Visualizing Decision Evidences

The final phase of the proposed framework is **Visualizing**. By applying the proposed Algorithm (1), it is possible to extract critical threat factors. We begin by matching the extracted critical threat factors with the original payload and then apply **Highlighting** to the payload.

$$\alpha_i = \min_opacity + \left(\frac{\max_opacity - \min_opacity}{n - 1} \right) \times i \quad (8)$$

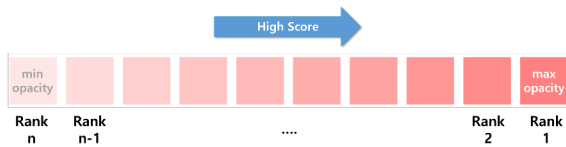


Fig. 6: Ranking based Highlight (min : 0.1, max : 0.5, N : 10)

To enhance the visibility of the results, we calculate calculated the ranking based on highlight **Coloring**. As shown in Figure 6 and Equation (8), where n represents count of highlight colors and i means top i_{th} rank.

In this way, through a framework that integrates the four phases, as shown in Figure 1, we provide SOC analysts with more suitable and clear interpretive information. This approach enables analysts to effectively support decision-making for detected suspicious data, enhances the reliability of AI models

in network security monitoring, and improves the efficiency of developing rapid response strategies for attacks.

IV. EXPERIMENT AND EVALUATION

This section introduces the real-world web data detected as suspicious in SOC environments and validates the effectiveness of the proposed methods. For validation, we extract the top features from the global explanations for each Approach. Comparison of them with the tagging data which is real-world network traffic and annotated threat factors by security monitoring experts. In other words, by comparing the Hit-rate metrics which means similarity between the tagging data and the extracted top features. we evaluate whether the proposed technique can effectively extract the factors that security analyst actually judge to be threat.

A. Dataset Preparation with Real-world Traffic

The data used in our experiments are Real-world Traffic data detected by the Network Intrusion Detection System (NIDS) rules. These NIDS rules were developed by the NCSC, and Korean public institutions operate NIDSs based on the above rules. These detected data were labeled with two categories by comparing the similarity of the payloads to actual security incident data from the real-world. Therefore, if the similarity is high, it is labeled as True Positive (1), and if the similarity is low, it is labeled as False Positive (0). In particular, in this experiment, we conducted experiments only on security events that can be converted to ASCII, i.e., data that is understandable to humans, to verify the proposed framework. These experimental data consist of web data collected over a total of 12 months in 2021 and 2022. The data distribution is described in Table I.

TABLE I: Dataset Statistics

Data set		False positive	True positive	Total
Malicious WEB Request	Training	55,274	23,461	78,735
	Test	23,689	10,055	33,744
Detected at 2021	Total	78,963	33,516	112,479
Malicious WEB Request	Training	66,254	14,351	80,605
	Test	28,395	6,150	34,545
Detected at 2022	Total	94,649	20,501	115,150

Figure 7 is example of our Real-world Traffic data. Example 7b has three annotated threat factors includes **'Itemlookup.do'** in the Request_URL, **'cn=/etc/passwd'** in the Request_Query, and **'Java/1.7.0_72'** in the User-Agent. This request indicates a malicious attempt to query the system file containing user account information using an automated script based on a Java client. Thus, security monitoring experts assigned three tagging information to summarize such activity.

B. Experiment

1) *Comparison test for extracting features*: First comparison of experiment is Extraction Semantic Features. The results of extracting each feature using the different methods, as shown in Table II.

Tagging	NIDS-WEB-Signature_\$\$\$\$
<pre>GET /wp-content/plugins/js_composer/assets/%252e%252e/%252e%252e/%252e%252e/%252e%252e/%252e%252e/etc/passwd?ver=4.3.3 HTTP/1.1 User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko Host: #####.ac.kr Accept: */* Connection: keep-alive</pre>	

(a) Malicious WEB Request Detected at 2021

Tagging	NIDS-WEB-Signature_####
<pre>GET /itemlookup.do?keyValue=09718604&target=JAKO&returnType=xml& responseGroup=advance&cn=/etc/passwd HTTP/1.1 User-Agent: Java/1.7.0_72 Host: #####.ac.kr Accept: text/html, image/gif, image/jpeg, */*; q=.2, */*; q=.2 Connection: keep-alive</pre>	

(b) Malicious WEB Request Detected at 2022

Fig. 7: Example of Detected SOC WEB Data & Tagging

TABLE II: Result of Extracting Features - '22 WEB

	Before	After
1	M: ing	U: /servlet/Linking
2	M: callb	Q: callback=
3	M: cord.jsp?cn=	Q: cn=/etc/passwd
4	M: la/5.0	A: Mozilla/5.0
5	M: eCo	Q: pageCode=PG04&
6	M: _name	Q: file_name=../..

Applying the proposed method, unlike the traditional approach, allows us to identify which specific category each feature was extracted from based on the initials of the four categories, such as $Q(request_Query)$:, $U(request_Url)$:, $R(Referer)$:, and $A(Agent)$:. Additionally, in traditional tokenization, features are extracted with mixed categories, including delimiters such as '?', which differentiate between Request_Query and Request_URL. In contrast, the proposed method is applied, each payload is split by category before tokenization, allowing tokenization to be focused exclusively on each segmented category.

Like this, compared to the previous, the proposed technique can extract more understandable and more semantic features related to network behavior.

2) *Re-ordering result by ranking risk influences*: The final experiment is comparison of Ranking results by each approach. As shown in Figure 8b, 8d the proposed method, unlike traditional methods, ranks only critical threat factors related to attacks at the top. This result reflects the effect of the proposed method that considers both consistency and actual influence.

As shown in Table III, we calculated the scores for each approach for the top features to compare the appropriateness of the proposed practical influence-based ranking. Especially, according to the experimental results, new features that were

TABLE III: Result of Ranking Process - '22 WEB

	Feature	Before	After
1	Q: cn=/etc/passwd	1.92	12.56
2	U: /tmui/lo	1.51	11.02
3	Q: table	2.3	10.43
4	U: fields	0.08	8.12
5	U: %uff0e%uff0e/	0.03	6.1
6	A: Java/1.	0.01	4.72
7	U: leFullRecord.js	0.25	3.19
8	Q: O	0.35	1.92
9	U: down.php	0.34	1.62
10	A: _Chrome/103.0	0.03	1.37

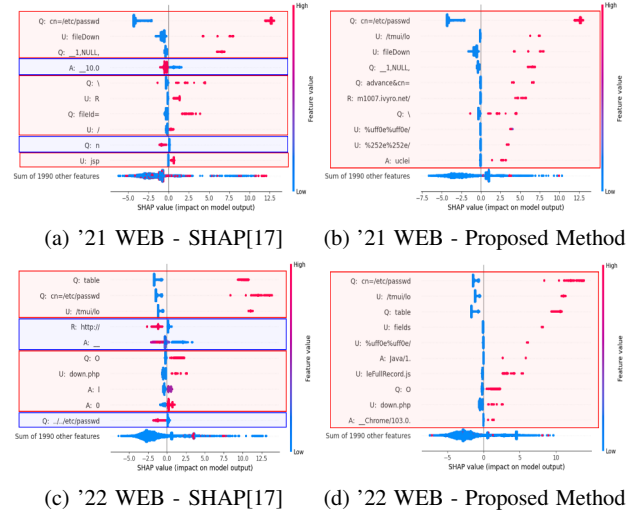


Fig. 8: Ranking Result between SHAP and Proposed Method

not detected by the traditional method appeared in the top 4th ~ 7th and 10th through proposed method.

Examining the Prop_Score of these features suggests that their high consistency and practical influence have resulted in their higher rankings. Conversely, the Trad_Score of these features was calculated to be low. This is because traditional ranking methods involve all data in the score calculation. Even if a feature has a high influence, if it has fewer attack cases, its overall average score decreases significantly, making it difficult to detect in the global top. In contrast, the proposed method measures practical influence more accurately by only considering instances where the feature is mapped, thereby focusing on its actual impact. As a result, the top-ranked features are consistently involved in attacks, and even features with a minority data group are identified at the top if they have a high importance for the attack.

C. Evaluation

1) *Explainability by hit-rate*: In evaluation section, we calculate and compare the Hit_Rate between the traditional method and the proposed method to numerically evaluate the appropriateness of the proposed approach. To calculate the Hit-rate, which means the similarity between the extracted features and the tagging data, the tagging data was segmented into the

(a) Malicious WEB Request Detected at 2021

	Labeled Annotation	Count	Trad.	Prop.
1	(U) UNION ALL SELECT 1, NULL,script Ealert(XSS)...	4268	4268	4268
2	(U) /tmui/login.jsp/.../locallb/... (Q) /etc/passwd	4200	2100	2100
3	(U) /simple-fields/simple_fields.php (Q) etc/passwd%00	213	0	107
4	(U) /article/ArticleFullRecord.jsp (Q) cn=/etc/passwd (R) /article/ArticleFullRecord..	113	37	75
5	(U) Itemlookup.do (Q) cn=/etc/passwd (A) Java/1.7.0.72	9	3	6
6	(U) /%uff0e%uff0e	7	7	7
7	Others	109	-	-
Total Hit-rate (%)			47.12	72.81

Labeled Annotation	Count	Trad.	Prop.
UNION ALL SELECT	4268	4268	4268
NULL.script Ealert(XSS)...			
/tmui/login.jsp/././localb/...	4200	2100	2100
/etc/passwd			
/simple-fields/simple_fields.php	213	0	107
etc/passwd%00			
/article/ArticleFullRecord.jsp			
cn=/etc/passwd	113	37	75
/article/ArticleFullRecord..			
Itemlookup.do			
cn=/etc/passwd	9	3	6
Java/1.7.0.72			
/uff0e%uff0e	7	7	7
thers	109	-	-
Total Hit-rate (%)		47.12	72.81

Overall, when analyzing the results, a comparison of similarity across each approach for the six data groups shows that Groups 1 and 2, which have a larger amount of data, exhibit similar similarity scores. In contrast, Groups 3 through 6, which have less data, demonstrate the highest similarity scores with the proposed methods. It shows that the proposed technique effectively extracts significant features, even for less prevalent categories that were challenging to detect from a global perspective.

(a) Malicious WEB Request Detected at 2021

(b) Malicious WEB Request Detected at 2022

2) *Visualization examples:* In the final step, we calculate the ranking-based highlight colors for the top N extracted features. As shown in Figure 9, we conduct matching and highlighting of the original payload based on the top 10 features and their corresponding highlight colors.

Finally, we extracted critical threat factors based on network behaviors and conducted an experiment to present these factors in a more suitable form to security analysts in the SOC environment, thereby providing decision support.

In this research, we specifically focus on providing more suitable and clear interpretative information in real-world SOC environments, proposing practical XAI techniques to support decision-making. The proposed framework extracts critical threat factors from raw payloads by XAI and visualizes them in a more intuitive form for real-world SOC environments.

However, this experiment was conducted with a focus on security events that could be converted to ASCII for verification purposes. Therefore, the current framework has a limitation in that it focuses on data that can be converted into

an understandable payload. To overcome this and generalize the application of the proposed framework, it is necessary to verify its suitability by conducting experiments based on various network data.

Additionally, the current experiment validated the relevance of the extracted features based on metrics such as the Hitting_Rate. In future research, we going to develop various performance metrics, such as an index of improved work efficiency of security analysts in SOC environments through the proposed method. In future research, we going to develop various performance metrics, such as an metrics of improved work efficiency of security analysts in SOC and indicators of efficiency improvement and semantic preservation scores. This future work aims to verify the suitability of the proposed method.

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